

Symbol	Symbol Name
$\{\}$	set
$A \cup B$	union
$A \cap B$	intersection
$A \subseteq B$	subset
$A \not\subseteq B$	not subset
$A \subset B$	proper subset / strict subset
$A \supset B$	proper superset / strict superset
$A \supseteq B$	superset
$\emptyset$	empty set
$P(C)$	power set
$A \not\supset B$	not superset

$A \not\supset B$	not superset
$A = B$	equality
$A \setminus B$ or $A - B$	relative complement
A^c	complement
$A \Delta B$	symmetric difference
$a \in B$	element of
(a, b)	ordered pair
$x \notin A$	not element of
$ B $	cardinality
$A \times B$	cartesian product
N_1	natural numbers / whole number
N_0	natural numbers / whole number
Q	rational numbers set

Universal Set

A universal set is usually denoted by the capital letter 'U'. Also, sometimes it is denoted by ϵ (epsilon). It is a set that contains all the elements of other sets including its own elements.

$U = \{\text{counting numbers}\}$

$U = \text{Set of integers}$

Complement of Set

If A is a set, then the complement of set A will contain all the elements in the given universal set (U), that are not in set A. It is usually denoted by A' or A^c .

$A' = \{x \in U : x \notin A\}$

1. Let A and B be two finite sets such that $n(A) = 20$, $n(B) = 28$ and $n(A \cup B) = 36$, find $n(A \cap B)$.

Solution: Since, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

$$\text{So, } n(A \cap B) = n(A) + n(B) - n(A \cup B)$$

$$= 20 + 28 - 36$$

$$= 48 - 36$$

$$= 12$$

2. Let $A = \{x : x \text{ is a natural number and a factor of } 18\}$ and $B = \{x : x \text{ is a natural number and less than } 6\}$. Find $A \cup B$.

Solution: Given,

$$A = \{1, 2, 3, 6, 9, 18\}$$

$$B = \{1, 2, 3, 4, 5\}$$

$$\text{Therefore, } A \cup B = \{1, 2, 3, 4, 5, 6, 9, 18\}$$

3. Let $A = \{3, 5, 7\}$, $B = \{2, 3, 4, 6\}$. Find $(A \cap B)'$.

Solution: Given, $A = \{3, 5, 7\}$, $B = \{2, 3, 4, 6\}$

$$A \cap B = \{3\}$$

Therefore,

$$(A \cap B)' = \{2, 4, 5, 6, 7\}$$

4. If $A = \{2, 3, 4, 5, 6, 7\}$ and $B = \{3, 5, 7, 9, 11, 13\}$, then find (i) $A - B$ and (ii) $B - A$.

Solution: Given,

$$A = \{2, 3, 4, 5, 6, 7\} \text{ and } B = \{3, 5, 7, 9, 11, 13\}$$

$$(i) A - B = \{2, 4, 6\}$$

$$(ii) B - A = \{9, 11, 13\}$$

Practice Questions on Set Theory Symbols

1. If $A = \{1, 3, 5, 7\}$ and $B = \{2, 3, 4, 5\}$, then find $A \cup B$.
2. Let $P = \{a, e, i, o, u\}$ and $Q = \{a, b, c, d, e\}$. Find $P \cap Q$.
3. If $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 5, 7, 9, 11\}$, then find $A - B$.